

PhaseShift Technologies — Venture Brief (provisional, 5 pages)

Global Startup Campus / SRI Venture Build Program — application attachment Version 0.1, September 4, 2026. Pre-incorporation team. Contact: Junichi Kato, junichi3112btm@gmail.com, +81-80-3301-0725.

Provisional 5-page format. Figures follow our business plan v1.9 (Aug 29, 2026) and the price revision of Sep 1, 2026. "Estimate" = our calculation from public data; "not yet measured" = open. Customer names are withheld (Producer A, Producer B) because those discussions are confidential.

1. Summary

PhaseShift recovers iodine from the dilute wastewater that today's iodine plants discharge, using a temperature-responsive polymer that captures iodine at single-digit ppm and releases it on a temperature swing. The unit is a side-stream retrofit that the producer owns; we earn recurring revenue from polymer supply and a per-kilogram process fee on the iodine recovered.

Origin	Doshisha University (Kyoto), Faculty of Science and Engineering. PI: Assoc. Prof. Shinnosuke Nishimura (inventor).
Stage	Laboratory proof-of-concept, batch scale (TRL3). TRL4 assessment on real wastewater from two domestic producers scheduled for late September 2026.
Beachhead	Iodine producers on the Boso gas field, Chiba, Japan (about 80% of Japan's output; Japan supplies about 30% of the world's iodine).
Global path	Produced-water operators in Oklahoma's Anadarko Basin (Phase 2, gated on domestic results).
IP	Foundational polymer patent application published (Doshisha University). First application patent scheduled for filing October 2026.
Team	Business lead: Junichi Kato (five foundings, cumulative JPY 450M seed raised). Technical Advisor / PI: Assoc. Prof. Nishimura. One research staff member.
Funding plan	Two-stage J-KISS (JPY 70M angel, JPY 120M seed). JST D-Global 3-year R&D application due Sep 30, 2026.
Ask of GSC	US customer discovery in produced water, business-model stress test, US/PCT IP strategy, investor readiness, third-party validation partners.

2. The problem: iodine that is produced but never collected

Japan is the world's second-largest iodine producer. The iodine comes from natural-gas brine pumped in Chiba Prefecture and is recovered by two commercial methods, blow-out and ion exchange, which work at raw-brine concentrations (tens of ppm to about 100 ppm) and become uneconomic below that.

From published operating data for a representative plant (35 ppm brine, 60,000 t/day, 1,400 kg/day of iodine), we estimate that only about two-thirds of the iodine entering the plant is recovered. The rest, roughly 700 kg/day at that plant and, by our estimate, **700–1,000 t/yr nationwide**, leaves with the discharged brine at 7–12 ppm. One major producer has told us its discharge carries about 10 ppm, consistent with our estimate. (The national figure is being cross-checked with the Society of Iodine Science; until then it is our estimate.)

Producers cannot pump more brine: pumping is capped by ground-subsidence agreements with local governments. **Recovering iodine from their own discharge is therefore one of the few ways left for a Japanese producer to increase output.** That is what we sell: more iodine from the same brine, with the existing process untouched.

Where iodine goes, and why demand is not the constraint. By volume, iodine is used in X-ray contrast media, pharmaceuticals, LCD polarizing film, iodophor disinfectants, animal feed and fluorochemicals, in that order (USGS). Spot prices averaged USD 70/kg in 2024 and USD 73/kg over January–October 2025 (USGS). Two newer uses add a supply-security dimension. Japanese industrial policy names iodine a principal raw material for perovskite solar cells, which Japan intends to manufacture domestically at scale. In space, iodine is displacing xenon as an electric-propulsion propellant: the first in-orbit demonstration of an iodine thruster was published in *Nature* in 2021, and iodine stores as an unpressurised solid at roughly three times xenon's density, which simplifies small-satellite systems. Xenon output is concentrated (Russia and Ukraine, about a quarter to a third of world supply) while third-party projections put more than 20,000 satellites in orbit within a decade, most requiring electric propulsion. **Propulsion volumes are small today against a roughly 34,000 t/yr iodine market, and we put none of this in our financials.** The point is directional: two

industries that matter to both governments now draw on the same constrained primary supply, and Japan is the second-largest source of it.

3. Technology and evidence

3.1 How it works

The core material is a temperature-responsive polymer (the "PhaseShift agent") from the Nishimura laboratory, Doshisha University. In water it complexes with iodine and, on a modest temperature change, phase-separates into a concentrated iodine-rich phase; a further temperature swing releases the iodine. Desorption is therefore **designed** to need no reducing agents or solvents, and the heat source is **designed** to be low-grade plant waste heat (one producer has disclosed a steam-drain stream at 80–90 °C). Only the concentrated phase, about 1–3% of the flow, is heated; we estimate about 10.5 kW per unit. Whether waste heat alone suffices depends on heat-exchanger efficiency and is not yet demonstrated.

Installation is a **side-stream retrofit**: 3.5–6.8% of the discharge is diverted through a fixed-bed cartridge unit and returned; the main line never stops. This followed producer feedback that full-flow insertion was not feasible.

3.2 What has been measured (laboratory, batch, as of August 2026)

System	Conventional polymer (PVP)	PhaseShift agent	Note
Iodine in hexane	73.9%	91.4%	capture
Pure water, 7 ppm (two conditions)	—	96.8% / 97.2%	capture, UV-vis 351 nm, supernatant basis
Model brine wastewater, 7 ppm	—	supernatant below detection limit	capture
Reuse, second cycle	—	79.4%	dose not recorded; cause under investigation

These are **capture** figures derived from the drop in supernatant absorbance. They do not include desorption and collection. **Recovery on real wastewater has not yet been measured.** Our business plan uses a recovery rate placeholder of 92.9%, which we flag wherever it drives a number.

3.3 Known gaps (disclosed at application)

1. **Polymer loss of about 30% per cycle** (up to 50%) in the current liquid-liquid separation, because the dilute phase retains dissolved polymer; this bears directly on consumables economics. A copolymer that precipitates as a solid (gravity settling instead of centrifugation) has been made; its loss rate is not yet measured.
2. **No continuous-flow data.** The lab protocol uses static settling and centrifugation; a producer has named continuous flow its heaviest requirement. A continuous-flow unit is the first engineering task.
3. **Iodine speciation.** Discharge may carry residual reducing agent, leaving iodide rather than iodine; pre-oxidation requirements are being defined.
4. **High salinity.** Phase-separation temperature shifts with salt (38 °C in water, 35 °C in brine). Behavior above 100,000 mg/L TDS, typical of US produced water, is untested.

4. Market and customers

4.1 Phase 1 — Japan (beachhead)

Item	Value	Basis
Uncaptured iodine, national	700–1,000 t/yr	estimate from published plant data; being cross-checked
Value at our net price (JPY 8,587/kg)	JPY 6.0–8.6 bn/yr	net-to-gross ratio under review
Addressable by our model (Chiba, top-5 municipalities)	JPY 2.95–4.21 bn/yr	Chiba 82% of national output × 59.8% in top-5 municipalities
Per site (design case)	4,316 kg/yr recovered; JPY 50.9M/yr additional sales for the customer	2,074 m³/day side stream, 320 days, 7 ppm, recovery placeholder 92.9%, gross price JPY 11,782/kg (USD 74.27 × 158.64)
Sites by FY2030	Base 5 (downside 3, upside 7–9)	business plan v1.9

Customers are the producers themselves. Two of Japan's major producers are in technical discussions with us: Producer A under a three-party NDA since August 2026, with real wastewater samples agreed; Producer B since a first technical meeting on Aug 27, 2026, with samples agreed and a written list of adoption requirements (selective adsorption in brine, easy desorption, continuous-flow configuration). Chiba Prefecture has arranged a first meeting with a third producer for mid-September. One producer reports discharge of up to 18,000–19,000 m³/day, about nine times one unit's 2,074 m³/day design flow.

4.2 Phase 2 — North America (conditional)

US iodine comes largely from oil-and-gas produced water in Oklahoma's Anadarko Basin, where Iofina operates eight plants (each 10,000–50,000 bbl/day) and has announced a Permian plant for late 2026. Back-calculating from disclosed output, we estimate these plants take water at **59–108 ppm**. Two customer groups follow:

- **Downstream of existing plants.** If plant recovery is 70–90%, residual streams run 6–33 ppm, comparable to or richer than Japan's tails.
- **Streams that never reach a plant.** Water-midstream and saltwater-disposal (SWD) operators aggregate produced water (one operator alone reports 60 disposal facilities and 1.2 million bbl/day of capacity). Below the concentration at which iodine value equals disposal cost (**51–106 ppm** at USD 74.27/kg and disposal costs of USD 0.60–1.25/bbl), no one builds a dedicated plant. A low-CAPEX retrofit owned by the water operator is the only form that works there.

One Japanese design site (about 13,000 bbl/day) sits at the lower end of a US plant intake, so the unit transfers without redesign. Two open questions decide Phase 2 and are what we want to test through GSC: **do low-concentration streams exist at meaningful volume at aggregation points**, and **does the polymer work above 100,000 mg/L TDS**. If either answer is no, Phase 2 stops. Iodine is not on the US critical minerals list; we do not build the case on subsidies.

5. Business model and unit economics

Three revenue layers, all keyed to a unit the customer owns and operates:

Layer	Price (Sep 2026)	Nature
1. Unit sale	JPY 100M per unit (about USD 0.63M)	one-time; customer CAPEX
2. Consumables and maintenance	JPY 10M per year per unit	recurring polymer supply
3. Process fee on recovered iodine	JPY 2,000 per kg-I ₂	recurring, volume-linked

Customer P&L per site, per year (design case): additional iodine sales +JPY 50.85M; consumables –JPY 10.00M; process fee –JPY 8.63M; on-site power –JPY 0.60M; **net additional gross profit +JPY 31.62M** (gross margin 62%); simple payback **3.16 years** (about 2.2 years with Japanese investment tax incentives). All of these scale with the unverified recovery rate.

Why asset-light is a consequence, not a preference. Below the break-even concentration, iodine returns only part of the disposal cost, so a vertically integrated plant cannot pay for itself; the only design that works is a low-cost retrofit on existing assets, paid for by the asset owner and monetized through consumables and a volume fee.

An unfavorable number we disclose. Per kilogram of annual capacity our unit costs about USD 146 against about USD 44 for Iofina's newest plant, roughly 3.3×, because we process much thinner water. The gap narrows if North American tails are 20–30 ppm (3–4× output per unit) and is offset, in the customer's view, by the payback above.

Company revenue (illustrative, FY2030, 5 sites): about JPY 293M across the three layers. Cost of goods is being rebuilt after the September price revision, so we do not quote operating profit.

6. Competition and intellectual property

Alternative	Position	Our reading
Blow-out and ion exchange (incumbent)	Economical at higher concentrations; used by all Japanese producers	Neither side has published data at 7–12 ppm. We do not claim the band is uncontested.
Iofina IOsorb (USA)	Vertically integrated, 8 plants, 59–108 ppm feed (estimate)	We sit downstream, not in competition for the same water. Risk: Iofina moves to lower concentrations.
WO2025258648A1 (a Japanese university with a domestic producer)	Adsorbent for ≥10 ppm; desorption with reducing agent	Closest patent. Ours desorbs by temperature swing; regeneration data for both are limited.

Reverse osmosis	Rejection collapses at low feed (about 10% below 15 ppm in one study)	Not viable in our band.
Do nothing	Discharge or dispose	The real alternative. We add revenue rather than replacing disposal.

IP. A foundational patent application covering the polymer family is filed and published in Japan (Doshisha University, sole applicant; inventor: the PI). Our patent attorney has confirmed that a design outside the foundational claims is feasible; its structure is withheld until the **first application patent** (joint filing, Doshisha University and Kato; invention committee approval Aug 7, 2026) is filed in **October 2026**. Information is managed in three layers: public (function, targets, customer value), demonstration-required (operating conditions, analytics, connection specs) and core (formulation, manufacturing, control logic, cartridge internals). University-led public R&D results stay with the university under an exclusive option; company-led R&D is contracted so new IP is solely owned.

7. Global strategy and gates

Phase	Scope	Gate to proceed
1 (now–2028)	Japan: TRL4 → paid field unit at a producer	TRL4 result (Sep 2026); paid demonstration or equivalent LOI
2 (earliest 2027 Q2)	12-week North America feasibility study, then field demonstration	Nine conditions incl. domestic TRL4 pass, application patent filed, ring-fenced funding, a US prime partner candidate, a site candidate and a third-party verification lab

The order is fixed: North America is where a proven domestic design goes next, not a parallel bet. The GSC sprint sits between the two phases: it lets us answer the two Phase-2 questions and fill the partner-related gate conditions before committing money, without slowing domestic work.

8. Team and partners

Junichi Kato — Business lead / CEO candidate. Serial entrepreneur, five foundings. Built a Web3 solutions company in Malaysia to an 80-person multinational team; CEO of a Los Angeles-based satellite-data venture; cumulative JPY 450M seed raised. Since 2023, regional business development with Chiba local governments; Regional Revitalization Entrepreneur, Shirako Town. Full-time on PhaseShift since April 2026. University of Tokyo Matsuo-Iwasawa Lab AI Management Course; NEDO SSA.

Shinnosuke Nishimura, Ph.D. — Principal researcher; Technical Advisor to the venture. Associate Professor, Doshisha University. Inventor of the foundational patent. Has agreed to serve as PI for the JST D-Global application.

Research staff (1). Materials synthesis and process validation.

Support network. Doshisha startup-support coordinator (IP confirmation, contracts); patent attorney and INPIT specialist (neutral review); CIC / Chiba Prefecture startup program (monthly); Chiba Prefecture industry division (producer introductions); TIB STUDIO (Tokyo). Recruiting: CxO candidates for technology and finance, and a polymer manufacturing / unit engineering partner.

9. Milestones and funding

When	Milestone
Sep 2026	TRL4 assessment on real wastewater (Producer A, Producer B; samples agreed) with a predefined go / conditional-go / pivot / stop rule. JST D-Global application (3-year R&D, FY2027–29) with a commercialization-promotion organization.
Oct 2026	First application patent filed.
Oct 2026–Feb 2027	GSC sprint; Silicon Valley bootcamp Nov 30–Dec 11.
2027	100 L/day bench unit (TRL5): mass balance, consumable life, waste-heat energy balance.
2028	On-site pilot at a producer, side-stream (TRL6).
2029	North American sample validation and partner framework (TRL7); go / no-go on Phase 2 field demonstration.

Funding. Two-stage J-KISS: JPY 70M angel, then JPY 120M seed (JPY 190M total). Incorporation is tied to public-funding outcomes rather than a fixed date: if D-Global is awarded, R&D runs at the university first and the company is

formed at the end of that period; if not, immediately. An ERCA grant (JPY 4M) funds part of the 2026 work. Unit economics assume no subsidies.

10. What we want from GSC, and how we will know the sprint worked

1. **US customer discovery in produced water** — introductions to water-midstream / SWD operators and produced-water experts in Oklahoma and Texas. *Success*: a concentration-and-volume map of candidate aggregation points and two operators willing to share water-quality data.
2. **Business-model stress test** — mentors who have commercialized separation or water technologies, including offtake for concentrated iodine (US water operators, unlike Japanese producers, do not refine). *Success*: a US-specific model with named offtake options.
3. **IP and regulatory strategy** — US/PCT filing plan for the October 2026 application, freedom to operate versus IOsorb-type resins, produced-water rules. *Success*: a filing decision and an FTO memo.
4. **Investor readiness** — feedback from deep-tech and climate investors on the milestones that make a Japan-first resource-recovery venture fundable. *Success*: a seed narrative and target list for 2027.
5. **Third-party validation partners** — a US university or lab for high-salinity tests. *Success*: one signed test plan.

We bring real wastewater data from Japanese producers, a numbered gate framework, and a business lead who has built and raised in Southeast Asia and the US and works directly with mentors and investors in English.

11. Risks and open questions

Risk	Our response
TRL4 cannot be run in September or lands in the "hold" band	Quantification route for low ppm being secured (ion chromatography, external labs); decision timing slips to October rather than the plan being changed.
Recovery rate (placeholder 92.9%) comes in low	All site-level figures are recomputed; exit thresholds are pre-set and will not be loosened without new evidence.
Polymer loss stays near 30%; waste heat insufficient	Solid-precipitating copolymer, else consumables pricing and contract structure are redesigned. Reduce heated fraction, direct heat exchange at the separator, or auxiliary heat; if all fail, re-price on purchased power.
Incumbents raise their own recovery and tails shrink	Establish operating data and installed base first; bromine co-recovery from the same brine as a later option (not in the financials).
Polymer fails above 100,000 mg/L TDS	Phase 2 stops or shifts to lower-TDS streams.
Iofina moves down in concentration; separation mechanism alone is a weak barrier	Ownership model and installed data in Japan; downstream positioning in the US; barrier moved to operating data, installation know-how and consumable supply.

End of brief.